

Integral Cryptanalysis on the BAKSHEESH Cipher

3-Round Integral Trails

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Basic Set of Integral Cryptanalysis

Integral Cryptanalysis tracks a **structured multiset** of plaintexts.

Let the 128-bit BAKSHEESH state consist of 32 nibbles (4-bit words). We construct a set of $2^4 = 16$ chosen plaintexts:

$$P_i = (i, c_1, c_2, \dots, c_{31}) \quad \text{for } 0 \leq i \leq 15$$
$$\mathcal{P} = \{P_0, P_1, P_2, \dots, P_{15}\}$$

Multiset of chosen plaintexts of 16 Plaintexts

One nibble takes all values in $\{0, 1\}^4$, others are fixed constants $c_j \in \{0, 1\}^4$.

i	c_4	c_8	c_{12}	c_{16}	c_{20}	c_{24}	c_{28}
c_1	c_5	c_9	c_{13}	c_{17}	c_{21}	c_{25}	c_{29}
c_2	c_6	c_{10}	c_{14}	c_{18}	c_{22}	c_{26}	c_{30}
c_3	c_7	c_{11}	c_{15}	c_{19}	c_{23}	c_{27}	c_{31}

The Properties: $\mathcal{A}$ and $\mathcal{C}$

All / Active ($\mathcal{A}$)

A nibble where **all 16 possible values** appear exactly once across the multiset.

Constant ($\mathcal{C}$)

A nibble where all 16 texts share an **identical value**.

Initial plaintext set $\mathcal{P}$ mapped in terms of $\mathcal{A}$ and $\mathcal{C}$:

$\mathcal{A}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$

Processing $\mathcal{P}$ through Pre-Whitening (Key XOR)

Unlike GIFT-128, BAKSHEESH begins encryption with a full round key XOR. Let sk_0 be the 32-nibble master sub-key. We XOR sk_0 with all 16 states in $\mathcal{P}$:

$$P^{AK} = \{P_0 \oplus sk_0, P_1 \oplus sk_0, \dots, P_{15} \oplus sk_0\}$$

$i \oplus sk_0[0]$	$c_4 \oplus sk_0[4]$	...	$c_{28} \oplus sk_0[28]$	$\Rightarrow$	$\mathcal{A}$	$\mathcal{C}$	...	$\mathcal{C}$
$c_1 \oplus sk_0[1]$	$c_5 \oplus sk_0[5]$	...	$c_{29} \oplus sk_0[29]$		$\mathcal{C}$	$\mathcal{C}$	...	$\mathcal{C}$
$c_2 \oplus sk_0[2]$	$c_6 \oplus sk_0[6]$	...	$c_{30} \oplus sk_0[30]$		$\mathcal{C}$	$\mathcal{C}$	...	$\mathcal{C}$
$c_3 \oplus sk_0[3]$	$c_7 \oplus sk_0[7]$	...	$c_{31} \oplus sk_0[31]$		$\mathcal{C}$	$\mathcal{C}$	...	$\mathcal{C}$

Lemma

By XORing a constant to each text in the set, the nibble with the **all** ($\mathcal{A}$) property still satisfies the all property, and the **constant** ($\mathcal{C}$) property remains constant.

Round 1: Processing $\mathcal{P}$ through SubCells

The first operation in Round 1 is the **SubCells (SB)** layer, where the 4×4 SBox is applied independently to all 32 nibbles.

$$S_1^{SB} = \{SB(P_0^{AK}), SB(P_1^{AK}), \dots, SB(P_{15}^{AK})\}$$

$\mathcal{A}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$



$\mathcal{A}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$

Reasoning (Bijective Property)

Because the BAKSHEESH SBox is a **bijection**, the 16 distinct values in the $\mathcal{A}$ nibble remain distinct after substitution. Hence, $\mathcal{A} \rightarrow \mathcal{A}$. Constant nibbles also remain constant, so $\mathcal{C} \rightarrow \mathcal{C}$.

Round 1: Processing $\mathcal{P}$ through PermBits

BAKSHEESH uses the bit-permutation P_{128} . Tracing the 4 bits of the active nibble w_0 :

- Bit 0 $\rightarrow P_{128}(0) = \mathbf{0}$ (lands in nibble w_0)
- Bit 1 $\rightarrow P_{128}(1) = \mathbf{33}$ (lands in nibble w_8)
- Bit 2 $\rightarrow P_{128}(2) = \mathbf{66}$ (lands in nibble w_{16})
- Bit 3 $\rightarrow P_{128}(3) = \mathbf{99}$ (lands in nibble w_{24})

$\mathcal{A}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$

 $\Rightarrow$

$\mathcal{A}^*$	$\mathcal{C}$	$\mathcal{A}^*$	$\mathcal{C}$	$\mathcal{A}^*$	$\mathcal{C}$	$\mathcal{A}^*$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$

Bit Scattering Result

The active bits are scattered to 4 distinct nibbles. Each receives only 1 active bit, so they take 2 values across the set. They lose the strict $\mathcal{A}$ property and become $\mathcal{A}^*$. Their XOR sum remains 0, so they are balanced.

Round 1: AddConstants & AddRoundKey

The final steps of Round 1 are XORing the Round Constant (RC_1) and the Round 1 Sub-Key (sk_1).

$$S_1^{AK} = \{S_1^{PB} \oplus RC_1 \oplus sk_1\}$$

$\mathcal{A}^*$	$\mathcal{C}$	$\mathcal{A}^*$	$\mathcal{C}$	$\mathcal{A}^*$	$\mathcal{C}$	$\mathcal{A}^*$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$



$\mathcal{A}^*$	$\mathcal{C}$	$\mathcal{A}^*$	$\mathcal{C}$	$\mathcal{A}^*$	$\mathcal{C}$	$\mathcal{A}^*$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$

Reasoning (Linearity of XOR)

XORing any constant to a multiset does not destroy the properties of its component bits. The $\mathcal{A}^*$ nibbles (which are Balanced) stay Balanced, and the $\mathcal{C}$ nibbles stay Constant. The structural properties survive this step perfectly.

Balanced Property in Integral Cryptanalysis

Balanced Property ($\mathcal{B}$)

A nibble is called **Balanced** if the XOR sum of all its values across the multiset equals:

$$x_0 \oplus x_1 \oplus \cdots \oplus x_{15} = 0$$

Why Does Balance Appear After PermBits?

After the PermBits layer, each destination nibble receives only one active bit. Across the 16 plaintexts, that bit takes values:

$$0, 1, 0, 1, \dots$$

equally many times.

Therefore:

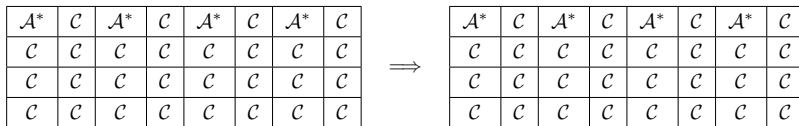
$$0 \oplus 1 \oplus 0 \oplus 1 \oplus \cdots = 0$$

Hence, the resulting nibbles become **Balanced** ($\mathcal{B}$).

Round 2: Processing $\mathcal{P}$ through SubCells

As Round 2 begins, the state contains 4 Partially Active ($\mathcal{A}^*$) nibbles and 28 Constant ($\mathcal{C}$) nibbles.

$$S_2^{SB} = \{SB(S_{1,0}^{AK}), \dots, SB(S_{1,15}^{AK})\}$$



Reasoning

SubCells mixes bits only inside each nibble.

Since the SBox does not diffuse across nibbles, the partially active nibbles remain partially active.

Round 2: PermBits & Key Addition

The PermBits layer scatters the bits of the 4 active nibbles across the state.

- $w_0 \rightarrow w_0, w_8, w_{16}, w_{24}$
- $w_8 \rightarrow w_2, w_{10}, w_{18}, w_{26}$
- $w_{16} \rightarrow w_4, w_{12}, w_{20}, w_{28}$
- $w_{24} \rightarrow w_6, w_{14}, w_{22}, w_{30}$

$\mathcal{A}^*$	$\mathcal{C}$	$\mathcal{A}^*$	$\mathcal{C}$	$\mathcal{A}^*$	$\mathcal{C}$	$\mathcal{A}^*$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$



$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$

Diffusion Result

The activity spreads into 16 nibbles whose XOR sums remain 0.

Hence, all affected nibbles become Balanced ($\mathcal{B}$). This property survives XOR with RC_2 and sk_2 .

Round 3: Processing $\mathcal{P}$ through SubCells

At the end of Round 2, the state contains:

16 $\mathcal{B}$ -nibbles and 16 $\mathcal{C}$ -nibbles

$$S_3^{SB} = \{SB(S_{2,0}^{AK}), \dots, SB(S_{2,15}^{AK})\}$$

$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$

$\Rightarrow$

$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$
$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$	$\mathcal{B}$
$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$

Observation

The SBox operates independently on each nibble and is bijective.

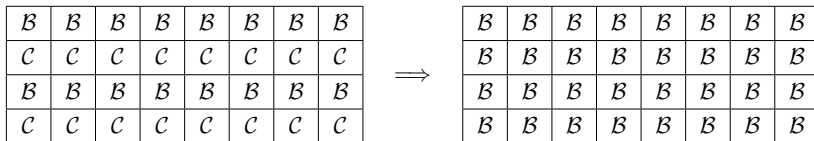
Therefore:

$$\mathcal{B} \rightarrow \mathcal{B}, \quad \mathcal{C} \rightarrow \mathcal{C}$$

Round 3: PermBits & Full Diffusion

The third PermBits layer scatters the bits of the 16 balanced nibbles across the entire state.

Because active information now exists in half of the state, the diffusion layer distributes activity to nearly all remaining positions.



3-Round Integral Property

After 3 rounds, every nibble of the state becomes Balanced:

$$\bigoplus_{i=0}^{15} X_i = 0$$

Thus, the attacker obtains a **3-round integral distinguisher** for BAKSHEESH.

What Does the Attacker Achieve?

The attacker prepares a structured multiset of:

$$2^4 = 16$$

chosen plaintexts with one active nibble.

After propagating through 3 rounds:

- Every nibble becomes Balanced.
- The XOR sum of each nibble equals 0.
- This behavior is highly non-random.

Why Is This Useful?

For a truly random permutation, such structured balance would occur only with negligible probability.

Therefore, the attacker can distinguish BAKSHEESH from a random permutation.

Main Result

A 3-round integral distinguisher is obtained for BAKSHEESH.

Complexity Analysis & Key Recovery

Complexity of the 3-Round Distinguisher

Metric	Requirement
Data (D)	$2^4 = 16$ Chosen Plaintexts
Time (T)	$\approx 2^4$ encryptions (XOR computation)
Memory (M)	Negligible (16 stored states)

Extending to an Attack (Key Recovery)

A distinguisher alone does not recover the key.

- **Append a Round:** Attack $R + 1$ rounds (e.g., 4 rounds).
- **Subkey Guessing:** Guess relevant final-round subkey bits.
- **Partial Decryption:** Decrypt ciphertexts backward by one round to the 3-round boundary.
- **Filtering:** If the XOR sum equals 0, retain the subkey guess as a likely candidate.

Practical Verification & Implementation

To empirically validate the theoretical model, a self-contained C++ framework was developed to simulate the integral property on BAKSHEESH.

Implementation Capabilities

- **Ground Truth:** Validates cipher components against the 5 official 35-round test vectors.
- **Dynamic Multiset:** Constructs the 16 chosen plaintexts for any active nibble.
- **Exhaustive Sweep:** Tests all 32 active nibble positions using 5 random 128-bit master keys.

Experimental Results

The implementation verifies that the XOR sum cancels to exactly 0 across all 32 output nibbles for all 160 trials. The distinguisher is fully key-independent.

Source Code & Documentation:

<https://github.com/skinnyduck21/CSL505-baksheesh-integral-cryptanalysis>